

Waves • Assessment Book

Six part-assessments plus a final mixed quiz, aligned to the presentation sequence and MATSEC 2027 Waves content.

How to use this book

Complete each part after that block of the topic. Each section mixes quick recall with explanation, calculations and practical thinking. Use **Print** for paper copies, or type directly into the boxes. The final quiz is a 20-question mixed check.

Suggested target: 70%+ on the multiple-choice checks before moving on; then correct written answers using teacher feedback.

Part 1 • Wave foundations

Sections 1–5: energy transfer, transverse/longitudinal waves, amplitude, wavelength, frequency, period, wave speed and graphs.

1. A wave transfers...

- ☐ A. matter only
- ☐ B. energy without overall transfer of matter
- ☐ C. particles from source to receiver
- ☐ D. mass and energy together

2. In a transverse wave, particle vibration is...

- ☐ A. parallel to energy transfer
- ☐ B. perpendicular to energy transfer
- ☐ C. always vertical
- ☐ D. in the same direction as wave speed

3. A wave has frequency 5 Hz. Its period is...

- ☐ A. 0.20 s
- ☐

☐ B. 5 s

☐ C. 2.5 s

☐ D. 25 s

4. A wave has $f = 8 \text{ Hz}$ and $\lambda = 0.50 \text{ m}$. Its speed is...

☐ A. 4 m/s

☐ B. 16 m/s

☐ C. 8.5 m/s

☐ D. 0.0625 m/s

5. Explain the difference between a displacement–distance graph and a displacement–time graph.

6. A wave travels at 12 m/s and has frequency 3 Hz. Calculate its wavelength. Show equation, substitution and unit.

7. Draw or describe one wavelength on a longitudinal wave.



Part 2 • Ripple tank & wave behaviour

Sections 6–9: wavefronts, stroboscope, reflection, refraction and diffraction.

1. A ray on a wave diagram is drawn...

- ☐ A. parallel to the wavefront
- ☐ B. perpendicular to the wavefront
- ☐ C. only for light
- ☐ D. along a crest

2. When water waves enter shallower water, their frequency...

- ☐ A. increases
- ☐ B. decreases
- ☐ C. stays the same
- ☐ D. becomes zero

3. Diffraction is strongest when the gap is...

- ☐ A. much wider than λ
- ☐ B. comparable to or smaller than λ

☐ C. always exactly 1 m

☐ D. at normal incidence

4. For reflection from a straight barrier, the angle of reflection is...

☐ A. twice i

☐ B. equal to i

☐ C. $90^\circ - i$

☐ D. always zero

5. Why is a stroboscope useful in a ripple-tank experiment?

6. Explain why wavelength decreases when a water wave enters shallower water while frequency stays constant.

7. Design a short investigation to test how gap width affects diffraction. Include one control variable and what you would observe.

Part 3 • Sound & ultrasound

Sections 10–12: sound as a mechanical longitudinal wave, speed, echoes, pitch, loudness and ultrasound.

1. Sound in air is...

- ☐ A. transverse electromagnetic
- ☐ B. longitudinal mechanical
- ☐ C. transverse mechanical
- ☐ D. longitudinal electromagnetic

2. Increasing sound frequency mainly increases...

- ☐ A. loudness
- ☐ B. pitch
- ☐ C. amplitude
- ☐ D. speed in the same medium

3. Ultrasound has frequency...

- ☐ A. below 20 Hz
- ☐

B. above 20 000 Hz

☐

C. exactly 340 Hz

☐

D. between 20 and 200 Hz

4. An echo returns after 0.40 s. Take speed of sound as 340 m/s. Find the distance to the wall.

5. Explain the bell-jar experiment evidence that sound needs a medium.

6. Distinguish pitch from loudness using wave quantities.

7. Explain how sonar can measure depth.

Part 4 • Electromagnetic spectrum

Section 13: common properties, order, uses, sources/detectors and visible light.

1. Which EM wave has the longest wavelength?

- ☐ A. gamma
- ☐ B. X-ray
- ☐ C. visible
- ☐ D. radio

2. All electromagnetic waves can...

- ☐ A. travel only in solids
- ☐ B. travel through a vacuum
- ☐ C. be heard by humans
- ☐ D. move slower than sound

3. As EM frequency increases, wavelength in vacuum...

- ☐ A. increases
- ☐ B. decreases

☐

C. stays constant

☐

D. becomes unrelated

4. Write the seven EM regions in order of increasing wavelength.

5. Give one use of infrared and explain the property that makes it useful.

6. An EM wave has frequency 1.0×10^8 Hz. Use $c = 3.0 \times 10^8$ m/s to calculate wavelength.

7. Put visible colours in order from longest to shortest wavelength.

Part 5 • Reflection, refraction & TIR

Sections 14–16: rays, plane mirrors, refraction, refractive index, apparent depth, critical angle and fibres.

1. A ray hits a mirror at 35° to the normal. Reflection angle is...

- ☐ A. 35°
- ☐ B. 55°
- ☐ C. 70°
- ☐ D. 145°

2. Light entering glass from air at an angle normally bends...

- ☐ A. away from the normal
- ☐ B. towards the normal
- ☐ C. along the boundary
- ☐ D. back to the source

3. At normal incidence, a ray entering glass...

- ☐ A. changes speed but not direction
- ☐

B. changes direction but not speed

☐

C. undergoes TIR

☐

D. must reflect completely

4. State two properties of an image in a plane mirror.

5. For a material, $\sin i = 0.75$ and $\sin r = 0.50$. Calculate refractive index $n = \sin i / \sin r$.

6. State both conditions needed for total internal reflection.

7. Explain why optical fibres can guide light around bends.

Part 6 • Dispersion & lenses

Section 17: dispersion, converging lenses, principal rays, image properties, magnification and focal length.

1. A converging lens is...

- ☐ A. thinner in the middle
- ☐ B. thicker in the middle
- ☐ C. always a plane mirror
- ☐ D. unable to form real images

2. For an object beyond $2F$, a converging lens forms an image that is...

- ☐ A. real, inverted, smaller
- ☐ B. virtual, upright, larger
- ☐ C. real, upright, same size
- ☐ D. virtual, inverted, smaller

3. For an object inside F , the image is...

- ☐ A. real and inverted
- ☐

B. virtual, upright and magnified

☐

C. real and same size

☐

D. at $2F$

4. Name the three principal rays commonly used to construct a converging-lens ray diagram.

5. An object is 4 cm high and its image is 10 cm high. Calculate magnification.

6. Why does a prism disperse white light?

7. Describe how to measure the focal length of a converging lens using a distant object.

Final assessment

20 questions

Final Waves Quiz

Do this without notes first. Then use your errors to choose what to revise.

1. Which statement best describes a wave?

- ☐ A. A transfer of energy without overall transfer of matter
- ☐ B. A transfer of matter only
- ☐ C. A permanent movement of particles with the wave
- ☐ D. A force that only exists in water

2. A particle vibrates parallel to the direction of energy transfer. The wave is...

- ☐ A. transverse
- ☐ B. longitudinal
- ☐ C. electromagnetic only
- ☐ D. stationary

3. A wave has $f=4$ Hz. Its period is...

- ☐ A. 0.25 s
- ☐ B. 4 s

☐
C. 2 s

☐
D. 8 s

4. A wave has $v=6$ m/s and $f=3$ Hz. Its wavelength is...

☐
A. 18 m

☐
B. 9 m

☐
C. 2 m

☐
D. 0.5 m

5. Water waves entering shallow water normally...

☐
A. speed up and λ increases

☐
B. slow down and λ decreases

☐
C. keep speed but change f

☐
D. stop at the boundary

6. Diffraction becomes stronger when...

☐
A. gap becomes large compared with λ

☐
B. gap becomes comparable with λ

☐

☐ C. frequency is always zero

☐ D. reflection angle increases

7. Sound cannot travel through...

☐ A. air

☐ B. water

☐ C. steel

☐ D. a vacuum

8. A larger sound-wave amplitude is mainly heard as...

☐ A. higher pitch

☐ B. greater loudness

☐ C. higher speed

☐ D. shorter period

9. An echo time is 0.60 s and sound speed is 340 m/s. Wall distance is...

☐ A. 204 m

☐ B. 102 m

☐ C. 340 m

- ☐
- D. 56.7 m

10. Ultrasound is sound above approximately...

- ☐
- A. 20 Hz
- ☐
- B. 200 Hz
- ☐
- C. 2 000 Hz
- ☐
- D. 20 000 Hz

11. Which order goes from longest to shortest wavelength?

- ☐
- A. radio → microwave → infrared → visible → ultraviolet → X-ray → gamma
- ☐
- B. gamma → X-ray → ultraviolet → visible → infrared → microwave → radio
- ☐
- C. radio → infrared → microwave → visible → gamma → X-ray → ultraviolet
- ☐
- D. visible → radio → gamma → infrared → X-ray → microwave → ultraviolet

12. All EM waves in vacuum have the same...

- ☐
- A. frequency
- ☐
- B. wavelength
- ☐
- C. speed
- ☐

D. energy

13. A mirror is hit at 28° to the normal. Reflection angle is...

☐ A. 28°

☐ B. 62°

☐ C. 56°

☐ D. 90°

14. A ray entering glass from air obliquely bends...

☐ A. towards the normal

☐ B. away from the normal

☐ C. only if it is red

☐ D. without changing speed

15. At normal incidence into glass, light...

☐ A. changes direction only

☐ B. changes speed but not direction

☐ C. undergoes TIR

☐ D. must stop

16. TIR requires light to travel...

- ☐ A. less dense to denser, any angle
- ☐ B. denser to less dense, $i >$ critical angle
- ☐ C. denser to less dense, $i <$ critical angle
- ☐ D. through a mirror

17. Optical fibres mainly rely on...

- ☐ A. diffraction
- ☐ B. dispersion
- ☐ C. total internal reflection
- ☐ D. sound reflection

18. White light separates in a prism because colours...

- ☐ A. all refract by exactly the same amount
- ☐ B. have different wavelengths and refract by different amounts
- ☐ C. have different speeds in vacuum
- ☐ D. are sound waves

19. Object beyond 2F in a converging lens gives...

- ☐ A. real inverted smaller image
- ☐ B. virtual upright larger image
- ☐ C. real upright image
- ☐ D. no image

20. Image height 12 cm, object height 4 cm. Magnification is...

- ☐ A. 0.33
- ☐ B. 3
- ☐ C. 8
- ☐ D. 48